Rumeysa Turkmen

PhD Candidate (ABD) in Computer Science | Concordia University
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OVERVIEW

My research is in Human-Computer Interaction, Virtual Reality, Augmented Reality, and 3D User Interfaces. Particularly, I design and evaluate 3D interaction techniques that make immersive environments more accessible, efficient, and usable, with a focus on eye-gaze, voice, and controller-free inputs for 3D selection and manipulation tasks. My broader work also includes 3D pointing, 3D sketching, and multimodal interaction in immersive environments. I am a recipient of the FRQNT doctoral research grant and was selected as a young researcher for the Heidelberg Laureate Forum.

EDUCATION

Doctor of Philosophy (PhD), Computer Science Fall 2024 – May 2027 (expected)
Concordia University *Montreal, QC, Canada*

- Direct-entry PhD admission based on undergraduate research publication record.
- Proposed dissertation title**:** *Investigating Controller-Free Interaction Techniques for Accurate 3D Selection and Manipulation Tasks.* Advisor: Dr. Anil Ufuk Batmaz.

Bachelor's Degree, Electrical and Electronics Engineering 2020 – 2024
Kadir Has University *Istanbul, Turkey*

- GPA: 3.47 / 4.00, graduated with an Honours Degree.
- Full scholarship recipient throughout undergraduate studies.

RESEARCH EXPERIENCE

Research Assistant — EXIT (EXtended Reality and Interaction Technologies) Lab Sep 2024 – Present
Concordia University *Montreal, QC, Canada*

- Conducting doctoral research on 3D interaction in Virtual and Augmented Reality.

Research Intern — TUBITAK-1002 Project Jul 2023 – Jun 2024
TUBITAK (Scientific and Technological Research Council of Turkey) *Turkey*

- Led an HCI/XR research project, “Input and Evaluation Methods of Eye-Gaze Behaviors to be Used on Virtual Reality Applications,” supervised by Dr. Mine Sarac Stroppa.
- Independently scoped the research direction, designed user studies, and contributed to multiple peer-reviewed publications.

Visiting Research Student — VERTEX Lab (HCI) Sep 2022 – Jan 2023
Dalhousie University *Halifax, NS, Canada*

- Conducted HCI research during undergrad studies, with Dr. Mayra Barrera Machuca, fully funded by the Study in Canada Scholarship.

Undergraduate Research Assistant — HapLab (Haptics) Sep 2021 – Jun 2024
Kadir Has University *Istanbul, Turkey*

- Researched eye-gaze behavior in 3D systems and haptic interaction with Dr. Mine Sarac Stroppa and Dr. Anil Ufuk Batmaz.

RESEARCH AND PUBLICATIONS

In Human-Computer Interaction and XR, conference proceedings are the preferred publication venue because they have the greatest impact. Top-tier conferences in this area conduct rigorous multi-stage peer review of full manuscripts and produce fully archival proceedings.

For first-author papers, I led the work by identifying and scoping the research direction, designing and developing the systems and study protocols, conducting user studies and statistical analyses, and writing the manuscripts. For non-first-author papers, I had an active role as a contributor and writer.

PEER-REVIEWED JOURNAL PAPERS

[J2] R. Turkmen, Z. Rasoulifar, A. K. Mutasim, and A. U. Batmaz. 2026. Investigating Depth-Based Target Expansion for Gaze Selection with Dwell Activation in Virtual Environments. To appear in *IEEE Transactions on Visualization and Computer Graphics **(TVCG)** (ISMAR Journal track).* 16% acceptance rate. (ICORE 2026: A*)

[J1] Z. Rasoulifar, **R. Turkmen**, M. D. Barrera-Machuca, W. Stuerzlinger, and A. U. Batmaz. 2026. Dynamic Vergence–Accommodation Conflict: Effects on 3D Selection Performance When Target Depth Changes. To appear in *IEEE Transactions on Visualization and Computer Graphics **(TVCG)**. (ISMAR Journal track)* 16% acceptance rate. (ICORE 2026: A*)

PEER-REVIEWED FULL CONFERENCE PAPERS

[C6] R. Turkmen, Z. Rasoulifar, and A. U. Batmaz. 2026. VoiceRay: 3D Object Selection Technique for Occluded and Dense Environments in Virtual Reality. *ACM CHI Conference on Human Factors in Computing Systems **(CHI '26)***. 25.3% acceptance rate. (ICORE 2026: A*)

[C5] R. Turkmen, L. Voisard, M. Kersten-Oertel, and A. U. Batmaz. 2025. Adaptive Hand Visibility for Accurate 3D User Interactions in Virtual Environments. *In Proceedings of the 2025 IEEE International Symposium on Mixed and Augmented Reality **(ISMAR '25)***. 20.6% acceptance rate. (ICORE 2026: A*)

[C4] R. Turkmen, F. R. Ortega, and A. U. Batmaz. 2025. Don't Miss Notifications: Exploring Gaze Notifications for Virtual Reality Cooking Environment. *In Proceedings of the 2025 IEEE International Symposium on Mixed and Augmented Reality **(ISMAR '25)***. 20.6% acceptance rate. (ICORE 2026: A*)

[C3] R. Turkmen, Z. E. Gelmez, A. U. Batmaz, W. Stuerzlinger, P. Asente, and M. Sarac. 2024. Eyeguide & Eyeconguide: Gaze-based Visual Guides to Improve 3D Sketching Systems. *In Proceedings of the 2024 ACM CHI Conference on Human Factors in Computing Systems **(CHI '24)***. 26.4% acceptance rate. (ICORE 2026: A*)

[C2] A. U. Batmaz, **R. Turkmen**, M. Sarac, M. D. Barrera Machuca, and W. Stuerzlinger. 2023. Effect of Grip Style on Peripersonal Target Pointing in VR Head-Mounted Displays. *In Proceedings of the 2023 IEEE International Symposium on Mixed and Augmented Reality **(ISMAR '23)***. 21.1% acceptance rate. (ICORE 2026: A*)

[C1] A. U. Batmaz, **R. Turkmen**, M. Sarac, M. D. Barrera Machuca, and W. Stuerzlinger. 2023. Re-investigating the Effect of the Vergence-Accommodation Conflict on 3D Pointing. *In Proceedings of the 29th ACM Symposium on Virtual Reality Software and Technology **(VRST '23)***. (ICORE 2023: A)

PEER-REVIEWED POSTERS AND WORKSHOP PAPERS

[A5] R. Turkmen and A. U. Batmaz. 2026. Cezve: A Virtual Turkish Coffee Pot Interaction Technique for 3D Target Selection in Virtual Reality. *In IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops **(IEEE VR '26)***.

[A4] R. Turkmen, N. Hatira, R. J. Teather, M. Kersten-Oertel, W. Stuerzlinger, et al. 2026. Investigating Adaptive Hand Visibilities for Accurate 3D User Interactions in Augmented Reality. *In IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops **(IEEE VR '26)***.

[A3] P. Rawat, **R. Turkmen**, C. Nwagu, K. Sunday, and M. D. B. Machuca. 2024. Evaluating Voxel-Based Graphical Passwords for Virtual Reality. *In IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops **(IEEE VR '24)***.

[A2] R. Turkmen, C. Nwagu, P. Rawat, P. Riddle, K. Sunday, and M. B. Machuca. 2023. Put Your Glasses On: A Voxel-based 3D Authentication System in VR Using Eye-Gaze. *In IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops **(IEEE VR '23)***.

[A1] R. Turkmen, K. Pfeuffer, M. D. Barrera Machuca, A. U. Batmaz, and H. Gellersen. 2022. Exploring Discrete Drawing Guides to Assist Users in Accurate Mid-Air Sketching in VR. *In Extended Abstracts of the 2022 ACM CHI Conference on Human Factors in Computing Systems **(CHI EA '22)***.

TEACHING & MENTORSHIP

Teaching Assistant 2024 – Present

Concordia University
COMP 376 — Introduction to Game Development
COMP 476 — Advanced Game Development

- Lead laboratory sessions and Unity tutorials to help students apply game development concepts.
- Support students with game development projects and answer questions about course concepts and implementation.
- Prepare instructional materials and tutorial resources for laboratory sessions.

Mentorship 2025 – 2026

Mentored a master's student who graduated in Summer 2026, providing guidance on experimental design, research methods, and the planning and execution of user studies. Supported manuscript preparation through feedback on paper structure, presentation of findings, and academic writing. This collaboration resulted in three co-authored papers published in *IEEE Transactions on Visualization and Computer Graphics (TVCG)* [J1, J2] and *ACM CHI* [C1].

AWARDS AND GRANTS

FRQNT Doctoral Research Grant (provincial) — CAD $100,000 May 2026 – May 2030
Fonds de recherche du Québec – Nature et technologies. Awarded through the Bourses de doctorat en recherche program for the doctoral research project "Sélection d'objets en VR grâce à la voix et à l'intelligence artificielle" (VR Object Selection Using Voice and Artificial Intelligence). DOI: 10.69777/2007432.

Gina Cody Scholarship (institutional) — CAD $60,000 2024
Concordia University, Gina Cody School of Engineering and Computer Science. Awarded to only 4 incoming PhD students per year based on academic excellence and research potential.

International Tuition Excellence Award (institutional) — CAD $47,000 2024 – 2027
Concordia University. Awarded to outstanding international PhD students to cover tuition differentials.

Study in Canada Scholarship (national) — CAD $10,000 2022
Government of Canada. Awarded to top international students to fund a research internship in Canada.

Full Undergraduate Tuition Scholarship (institutional) 2020 – 2024
Awarded based on national university entrance examination (YKS) ranking. One of 10 students out of 50 (top 20%) in the Electrical and Electronics Engineering cohort to receive a full tuition scholarship.

Second Best Design Project Award (institutional) 2024
Kadir Has University. Awarded to the runner-up graduation project across the Faculty of Engineering and Natural Sciences.

Honor Degree Graduate (institutional) 2024
Kadir Has University. Awarded to graduating students with sustained high academic standing.

TUBITAK Conference Travel Grant 2023

Concordia University Conference Travel Grant (Twice) 2026

RECOGNITIONS

Heidelberg Laureate Forum (HLF) Young Researcher (international) 2023
Selected as one of 200 young researchers worldwide for top early-career researchers in mathematics and computer science.

ACADEMIC SERVICE

Publicity Chair

- IEEE Conference on Virtual Reality and 3D User Interfaces (VR) 2027.
- ACM Symposium on Spatial User Interaction (SUI) 2026.
- ACM Symposium on Spatial User Interaction (SUI) 2025.
- ACM Symposium on Virtual Reality Software and Technology (VRST) 2025.

Peer Reviewer

- Reviewed 30+ papers for top HCI and XR venues, including International Journal of Human-Computer Studies, Computers & Graphics, ACM CHI, IEEE ISMAR, IEEE VR, ACM VRST, and ACM SUI.

Student Volunteer

- IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2023, 2025, 2026.

TALKS

- GirlSET Program, Concordia University: Demonstrated my VR/XR research and gave a research talk to high school students promoting women in science, engineering, and technology.
- MEDTEQ+: Presented our lab's work at a networking event organized by MEDTEQ+.
- Kadir Has University Outreach: Delivered talks to visiting high school students introducing my research in Virtual and Augmented Reality.

RESEARCH AREAS AND TECHNICAL SKILLS

Research Areas: Human-Computer Interaction, Virtual Reality, Augmented Reality, 3D User Interfaces, Eye-Gaze Interaction, 3D Selection and Pointing, Multimodal Input (voice, gaze, hand tracking), Haptic Interaction, Spatial Computing.

Technical: Unity (C#), VR/AR development for Meta Quest Pro and other HMDs, eye-tracking system integration, user study design and execution, quantitative and qualitative analysis, statistical methods (ANOVA, mixed models), Python, Java.

Languages: English (fluent), Turkish (native).